Deployment & Cost Plan

Clinical Intelligence Agent — Deployment & Cost Plan
Approach: **ECS + Fargate (Recommended)**
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Currency: USD (approximate; not an AWS quote)

Disclaimer: Costs are rough estimates for planning. Actual AWS, LLM, and Deepgram bills depend on region, usage, and configuration. Always set AWS Budgets and billing alerts.

1. Project Overview

The Clinical Intelligence Agent is a doctor-facing system that transcribes clinical conversations, labels doctor/patient roles, summarizes encounters, extracts entities, retrieves NICE guidelines via RAG, ranks differential diagnoses, and generates analysis + SOAP PDFs.

Current application tech stack (shared by both deployment approaches)

Layer	Technology	Role
API	FastAPI + Uvicorn	REST endpoints
Agent	LangGraph + MemorySaver	Stateful clinical pipeline
LLM	OpenRouter (e.g. DeepSeek) / optional Gemini/Groq	Summary, entities, diagnosis, insights
ASR	Deepgram Nova	Audio transcription + diarization
RAG	FAISS + HuggingFace MiniLM + metadata routing	Guideline retrieval
Observability	LangSmith	Tracing / monitoring
Reports	ReportLab PDFs	Analysis + SOAP notes
Frontend (planned)	Separate SPA (React/Next or similar)	Doctor UI
Auth (planned)	JWT / doctor user accounts	Session isolation

3. ECS + Fargate Architecture

- Frontend: S3 + CloudFront (static SPA) — separate deploy from backend
- Backend: ECS Fargate service behind Application Load Balancer (ALB)
- Images: Amazon ECR
- Secrets: AWS Secrets Manager (API keys, DB URL, JWT)
- Files: S3 (audio, PDFs, FAISS artifacts)
- Database (later): RDS Postgres for doctor users + visit ownership
- CI/CD: GitHub Actions — frontend pipeline ≠ backend pipeline
- Scaling: ECS Service Auto Scaling on CPU / ALB requests
- Optional later: SQS + second Fargate worker service for heavy analyze jobs

3.1 AWS tech stack for this approach

Component	AWS service	Purpose
Frontend hosting	S3 + CloudFront	Scalable static UI
Container registry	ECR	Backend (and optional FE) images

Component	AWS service	Purpose
Compute	ECS on Fargate	Run FastAPI tasks without managing EC2
Load balancing / TLS	ALB + ACM	HTTPS for api.domain.com
DNS	Route 53	app.* and api.* records
Config/secrets	Secrets Manager / SSM	Deepgram, OpenRouter, LangSmith keys
Object storage	S3	Audio, reports, guideline indexes
Database	RDS PostgreSQL	Users, sessions, visit metadata
Logs/metrics	CloudWatch	Ops monitoring
CI/CD	GitHub Actions → ECR/ECS/S3	Independent FE/BE releases

4. AWS Monthly Cost Estimates (ECS + Fargate)

Assumes region similar to us-east-1 / ap-south-1; low–moderate traffic.

AWS item	Staging (minimal)	Light prod (HA)
S3 + CloudFront (frontend)	$1–5	$3–15
ECR	~$1	$1–3
ALB	$16–25	$16–25
Fargate backend tasks	$25–45 (1 task)	$50–90 (2 tasks)
RDS Postgres (optional/small)	$0–15	$15–40
Secrets Manager	$1–3	$2–5
CloudWatch	$2–5	$5–15
NAT Gateway (if private)	$0–35	$30–35
AWS subtotal (typical)	**$45–90**	**$90–180**
With private NAT	**$75–120**	**$120–220**

2. External API Costs (LLM + Deepgram + LangSmith)

These costs apply to **both** EKS and ECS deployments. They often exceed AWS compute for AI apps.

2.1 Deepgram (Speech-to-Text)

Usage scenario	Assumptions	Rough monthly cost
Light demo	~50 consultations × 5 min = 250 min	~$5–15
Small clinic	~300 consults × 8 min = 2,400 min	~$30–80
Growing usage	~1,000 consults × 10 min = 10,000 min	~$100–300+

Note: Exact Deepgram pricing depends on model (Nova), features (diarization), and plan. Check current Deepgram pricing page.

2.2 LLM via OpenRouter / model APIs

Each analyze run may call the LLM multiple times: role labeling, summary, entities, RAG diagnosis, insights/SOAP.

Scenario	Assumptions	Rough monthly cost
Light demo	~50 full analyses; cheaper model (DeepSeek-class)	~$5–25
Small clinic	~300 analyses; mixed prompts	~$40–120
Heavier / stronger models	~300 analyses; higher-tier models	~$100–300+

Cost drivers: tokens in/out, number of LLM calls per visit, model choice, follow-up loops, PDF insight generation.

2.3 LangSmith

- Free/developer tiers may cover early demos
- Production tracing volume can add ~$0–50+/mo depending on plan and sampling
- Recommendation: sample traces in production (e.g. 10–20%) to control cost

2.4 Combined external API ballpark

Stage	Deepgram	LLM	LangSmith	API total
Demo	$5–15	$5–25	$0–10	**$10–50**
Small clinic	$30–80	$40–120	$0–30	**$70–230**
Growth	$100–300	$100–300	$20–50	**$220–650+**

5. Total Cost of Ownership (AWS + APIs)

Stage	AWS (ECS)	External APIs	Combined rough total
Demo / staging	$45–90	$10–50	**$55–140 / month**
Small clinic prod	$90–180	$70–230	**$160–410 / month**
Growth	$150–250	$220–650	**$370–900+ / month**

6. Scaling & Cost Optimization

- Keep frontend on S3/CloudFront — do not run static UI on Fargate unless required
- Start with 1 Fargate task for demos; use 2 for production HA
- Right-size task CPU/memory (RAG needs RAM more than vCPU)
- Later: move heavy analyze to SQS + worker service; scale workers on queue depth (even to 0)
- Use cheaper LLM for labeling/summary; stronger model only for diagnosis if needed
- Cache repeated analyses; sample LangSmith traces in production
- Stop staging services when not in use; set AWS Budget alerts day one
- S3 lifecycle rules for old audio/PDFs

7. Delivery Phases

- P1: Dockerize backend + health check
- P2: ECR + ECS Fargate service + ALB
- P3: S3/CloudFront frontend + separate CI/CD
- P4: Secrets Manager + S3 for files
- P5: RDS + doctor auth/session isolation
- P6: Autoscaling + optional async workers

Recommendation: Use ECS + Fargate for this project. Lower ops overhead and typically lower AWS cost than EKS for a FE/BE clinical agent.

Document type: ECS + Fargate deployment & cost plan — Clinical Intelligence Agent